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Exercise 4

SQl _JOINS

Question 1

```
85   -- Question 1:Show every user who has a subscription.Match users to subscriptions
86   SELECT DISTINCT u.User_id,
87   u.User_name,
88   s.Subscription_id,
89   s.Start_date
90   FROM exercise4.brightlearn.users u
91   INNER JOIN exercise4.brightlearn.subscriptions s ON u.User id = s.User id;
```

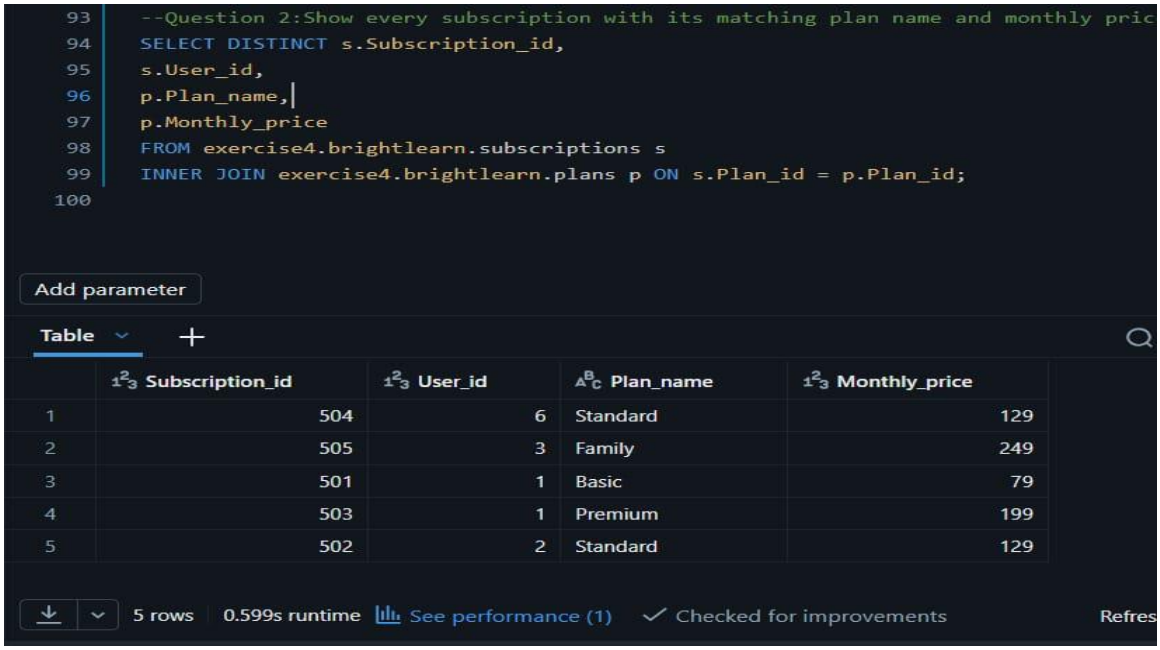
Add parameter

Table

	User_id	User_name	Subscription_id	Start_date
1	1	Nomvula	501	2026-01-15
2	1	Nomvula	503	2026-03-10
3	3	Anele	505	2026-04-05
4	2	David	502	2026-02-01

4 rows | 2.361s runtime | See performance (1) | Checked for improvements | Refreshed at

Question 2:



```
93  --Question 2:Show every subscription with its matching plan name and monthly pric
94  SELECT DISTINCT s.Subscription_id,
95  s.User_id,
96  p.Plan_name,
97  p.Monthly_price
98  FROM exercise4.brightlearn.subscriptions s
99  INNER JOIN exercise4.brightlearn.plans p ON s.Plan_id = p.Plan_id;
100
```

Add parameter

Table +

	Subscription_id	User_id	Plan_name	Monthly_price
1	504	6	Standard	129
2	505	3	Family	249
3	501	1	Basic	79
4	503	1	Premium	199
5	502	2	Standard	129

5 rows | 0.599s runtime | See performance (1) | Checked for improvements | Refres

Question 3 :

```sql
--Question 3: Show every viewing session that has a matching show.Include the show title and genre
SELECT DISTINCT v.Session_id,
       v.User_id,
       sh.Show_title,
       sh.Genre,
       v.Watch_Minutes
FROM exercise4.brightlearn.sessions v
INNER JOIN exercise4.brightlearn.shows sh ON v.Show_id = sh.Show_id;

--Question 4:Show every viewing session with the user who watched it . Only show sessions with matching u
```

Add parameter

Table

	Session_id	User_id	Show_title	Genre	Watch_Minutes
1	902	2	Tech Table	Documentary	30
2	901	1	Comedy Hour	Comedy	45
3	903	1	Crime Time	Drama	60
4	904	7	Comedy Hour	Comedy	20
5	905	3	Wild Earth	Documentary	90

5 rows | 1.186s runtime | See performance (1) | Checked for improvements | Refreshed at 10/

Question 4:

```sql
--Question 4:Show every viewing session with the user who watched it . Only show sessions with matching user
SELECT DISTINCT u.User_name,
       u.Country,
       v.Session_id,
       v.Watch_Minutes
FROM exercise4.brightlearn.users u
INNER JOIN exercise4.brightlearn.sessions v ON u.User_id = v.User_id;
```

Add parameter

Table

	User_name	Country	Session_id	Watch_Minutes
1	Anele	Durban	905	90
2	David	Cape Towm	902	30
3	Nomvula	Johannesburg	903	60
4	Nomvula	Johannesburg	901	45

4 rows 1.181s runtime See performance (1) Checked for improvements Refreshed at 10/1/26, 12:37 AM (Nov

Question 5 :

```sql
--Question 5:Show users along with their subscription plan name and the price . Use only users who have both a subscription and a valid plan
SELECT DISTINCT u.User_name,
       u.Country,
       p.Plan_name,
       p.Monthly_price,
       s.Start_date
FROM exercise4.brightlearn.users u
INNER JOIN exercise4.brightlearn.subscriptions s ON u.User_id = s.User_id
INNER JOIN exercise4.brightlearn.plans p ON s.Plan_id = p.Plan_id;

--Question 6:Show every user and any subscriptions they have . Users without subscriptions must still appear
```

Add parameter

Table

	User_name	Country	Plan_name	Monthly_price	Start_date
1	David	Cape Towm	Standard	129	2026-02-01
2	Nomvula	Johannesburg	Basic	79	2026-01-15
3	Nomvula	Johannesburg	Premium	199	2026-03-10
4	Anele	Durban	Family	249	2026-04-05

4 rows 0.879s runtime See performance (1) Checked for improvements Refreshed at 10/24/26, 3:04

Question 6:

```sql
--Question 6:Show every user and any subscriptions they have . Users without subscriptions must still
appear
SELECT DISTINCT u.User_id,
       u.User_name,
       s.Subscription_id,
       s.Start_date
FROM exercise4.brightlearn.users u
LEFT JOIN exercise4.brightlearn.subscriptions s ON u.User_id = s.User_id;
```

Add parameter

Table

	User_id	User_name	Subscription_id	Start_date
1	1	Nomvula	501	2026-01-15
2	1	Nomvula	503	2026-03-10
3	3	Anele	505	2026-04-05
4	2	David	502	2026-02-01
5	4	Kabelo	null	null
6	5	Lerato	null	null

6 rows | 0.959s runtime | See performance (1) | Checked for improvements | Refreshed at 10/1/26, 12:39 AM (N

Question 7:

```sql
--Question 7:Show every plan and the subscriptions on it .Plans with no subscribers must still appear
SELECT DISTINCT p.Plan_id,
       p.Plan_name,
       s.Subscription_id,
       s.User_id
FROM exercise4.brightlearn.plans p
LEFT JOIN exercise4.brightlearn.subscriptions s ON p.Plan_id = s.Plan_id;
```

Add parameter

Table

	Plan_id	Plan_name	Subscription_id	User_id
1	11	Standard	504	6
2	11	Standard	502	2
3	10	Basic	501	1
4	12	Premium	503	1
5	13	Family	505	3
6	14	Mobile	null	null

6 rows | 1.167s runtime | See performance (1) | Checked for improvements | Refreshed at 10/1/26, 12:40 AM

Question 8:

```sql
--Question 8:Show every show and any viewing session on it. Shows that were never watched must still appear
SELECT DISTINCT sh.Show_id,
       sh.Show_title,
       v.Session_id,
       v.Watch_Minutes
FROM exercise4.brightlearn.shows sh
LEFT JOIN exercise4.brightlearn.sessions v ON sh.show_id = v.Show_id;
```

Add parameter

Table

	Show_id	Show_title	Session_id	Watch_Minutes
1	704	Cooking Lab	null	null
2	701	Comedy Hour	904	20
3	703	Tech Table	902	30
4	705	Wild Earth	905	90
5	702	Crime Time	903	60
6	701	Comedy Hour	901	45

6 rows 0.733s runtime See performance (1) Checked for improvements Refreshed at 10/1/26, 12:41 AM (Now)

Question 9:

```sql
--Question 9:Show every viewing session and the user who watched it .Sessions referencing users that do not exist must still appear (with null user details)
SELECT DISTINCT v.Session_id,
       v.Show_id,
       v.Watch_Minutes,
       u.User_id,
       u.User_name
FROM exercise4.brightlearn.sessions v
LEFT JOIN exercise4.brightlearn.users u ON v.User_id = u.User_id
```

Add parameter

Table

	Session_id	Show_id	Watch_Minutes	User_id	User_name
1	904	701	20	null	null
2	903	702	60	1	Nomvula
3	905	705	90	3	Anele
4	902	703	30	2	David
5	901	701	45	1	Nomvula

5 rows | 0.678s runtime | See performance (1) | Checked for improvements | Refreshed at 10/1/26, 12:42 AM (Now)

Question 10:

```sql
--Question 10: Show every user , the plan they are on (if any), and the monthly price .Users without a subscription must still appear
SELECT DISTINCT u.User_name,
       u.country,
       p.Plan_name,
       p.Monthly_price
FROM exercise4.brightlearn.users u
LEFT JOIN exercise4.brightlearn.subscriptions s ON u.User_id = s.User_id
LEFT JOIN exercise4.brightlearn.plans p ON s.Plan_id = p.Plan_id;
```

Add parameter

Table

	User_name	country	Plan_name	Monthly_price
1	Lerato	Port Elizabeth	null	null
2	David	Cape Towm	Standard	129
3	Anele	Durban	Family	249
4	Kabelo	Pretoria	null	null
5	Nomvula	Johannesburg	Premium	199
6	Nomvula	Johannesburg	Basic	79

6 rows 1.457s runtime See performance (1) Checked for improvements Refreshed at 10/1/26, 12:42 AM (Nov

Question 11:

```sql
--Question 11: Show every user and every subscription , including users without subscriptions AND subscriptions referencing users that do not exist
SELECT DISTINCT u.User_id,
       u.User_name,
       s.Subscription_id,
       s.Start_date
FROM exercise4.brightlearn.users u
FULL OUTER JOIN exercise4.brightlearn.subscriptions s ON u.User_id = s.User_id;
```

Add parameter

Table

	User_id	User_name	Subscription_id	Start_date
1	1	Nomvula	501	2026-01-15
2	1	Nomvula	503	2026-03-10
3	3	Anele	505	2026-04-05
4	2	David	502	2026-02-01
5	4	Kabelo	null	null
6	5	Lerato	null	null

7 rows 1.314s runtime See performance (1) Checked for improvements Refreshed at 10/1/26, 12:43 AM (N

Question 12:

```sql
--Question 12: Show every plan and every subscription , including plans without subscribers AND
subscribers referencing plans that do not exist
SELECT DISTINCT p.Plan_id,
       p.Plan_name,
       s.Subscription_id,
       s.User_id
FROM exercise4.brightlearn.plans p
FULL OUTER JOIN exercise4.brightlearn.subscriptions s ON p.Plan_id = s.Plan_id;
```

Add parameter

Table

	Plan_id	Plan_name	Subscription_id	User_id
1	11	Standard	504	6
2	11	Standard	502	2
3	12	Premium	503	1
4	10	Basic	501	1
5	13	Family	505	3
6	14	Mobile	null	null

6 rows 1.368s runtime See performance (1) Checked for improvements Refreshed at 10/1/26, 12:44 AM

Question 13:

```sql
--Question 13: Show every show and every viewing session , including shows without sessions AND
sessionsreferencing shows that do not exist
SELECT DISTINCT s.Show_id,
       s.Show_title,
       v.Session_id,
       v.Watch_Minutes
FROM exercise4.brightlearn.shows s
FULL OUTER JOIN exercise4.brightlearn.sessions v ON s.Show_id = v.Show_id;
```

Add parameter

Table

	Show_id	Show_title	Session_id	Watch_Minutes
1	704	Cooking Lab	null	null
2	701	Comedy Hour	904	20
3	703	Tech Table	902	30
4	705	Wild Earth	905	90
5	702	Crime Time	903	60
6	701	Comedy Hour	901	45

6 rows | 1.333s runtime | See performance (1) | Checked for improvements | Refreshed at 10/1/26, 12:44 AM (No

Question

Question 14:

```sql
--Question 13: Show every show and every viewing session , including shows without sessions AND
sessionsreferencing shows that do not exist
SELECT DISTINCT s.Show_id,
       s.Show_title,
       v.Session_id,
       v.Watch_Minutes
FROM exercise4.brightlearn.shows s
FULL OUTER JOIN exercise4.brightlearn.sessions v ON s.Show_id = v.Show_id;
```

Add parameter

Table

	Show_id	Show_title	Session_id	Watch_Minutes
1	704	Cooking Lab	null	null
2	701	Comedy Hour	904	20
3	703	Tech Table	902	30
4	705	Wild Earth	905	90
5	702	Crime Time	903	60
6	701	Comedy Hour	901	45

6 rows 1.333s runtime See performance (1) Checked for improvements Refreshed at 10/1/26, 12:44 AM (No

Question 15

```sql
--Question 15: Show every plan and every subscription and every plan in one query - using FULL OUTER JOIN throughout.This is hardest question -get all gaps visible at once
SELECT DISTINCT u.User_id,
       u.User_name,
       s.Subscription_id,
       COALESCE(s.Plan_id, p.Plan_id) AS Plan_id,
       p.Plan_name
FROM exercise4.brightlearn.users u
FULL OUTER JOIN exercise4.brightlearn.subscriptions s ON u.user_id = s.user_id
FULL OUTER JOIN exercise4.brightlearn.plans p ON s.Plan_id = p.Plan_id;
```

Add parameter

Table

	User_id	User_name	Subscription_id	Plan_id	Plan_name
3	4	Kabelo	null	null	null
4	2	David	502	11	Standard
5	5	Lerato	null	null	null
6	null	null	null	14	Mobile
7	1	Nomvula	503	12	Premium
8	1	Nomvula	501	10	Basic

8 rows 1.283s runtime See performance (1) Checked for improvements Refreshed at 10/1/26, 12:46 AM